A Quick Introduction to Molecular Cell Biology

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Outline

Organisms Uni-Cellular or Multi- (forming tissues)

(cell) life $\sim$ networks of genes, DNA, RNA, proteins, small molecules

environmental signals $\rightarrow$ responses (e.g. gene expression)

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in multicellular organisms, also important:

- inter-cell communication
- overall regulation
- control of differentiation into various tissues

Prokaryotes, Eukaryotes, Archaea, and Viruses

- *eukaryotes*

cells w/ nucleus & compartments (mitochondria, . . .)
unicellular (e.g. yeast) or multicellular (e.g. plants,animals)

classified into algea/molds/protozoa, fungi, plants, animals

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cells w/o nucleus & well-defined compartments; mostly unicellular

  - *bacteria*

  - *archaea*: structure $\sim$ bacteria, gene expression $\sim$ eukaryotes

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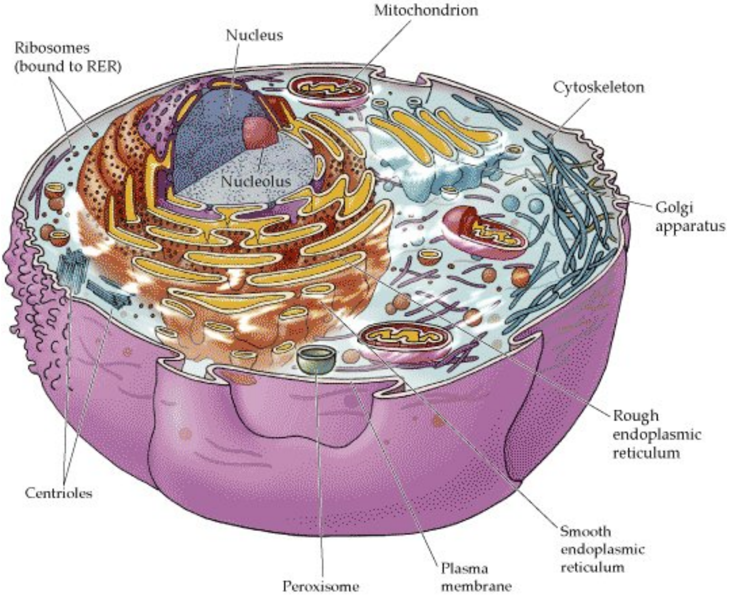
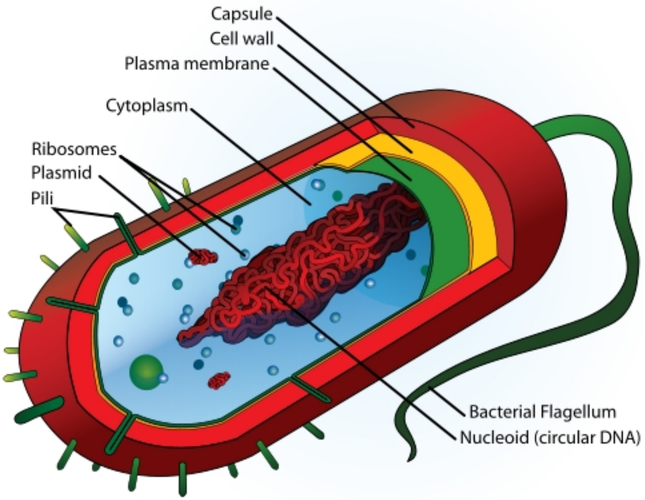
viruses: protein-coated DNA or RNA, not considered living
(cannot reproduce by themselves)

An Eukaryotic Cell



A Prokaryotic Cell



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Basic Principles Conserved in Evolution, Across Organisms

$\therefore$ biologists concentrate on certain *model systems*
(fruit flies, mice, yeast, worms, frogs)

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∴ biologists concentrate on certain *model systems*
(fruit flies, mice, yeast, worms, frogs)

- easier comparisons & sharing of research results

- aspects easier to study in different model organisms

by taking advantage of fast breeding or speed of maturation

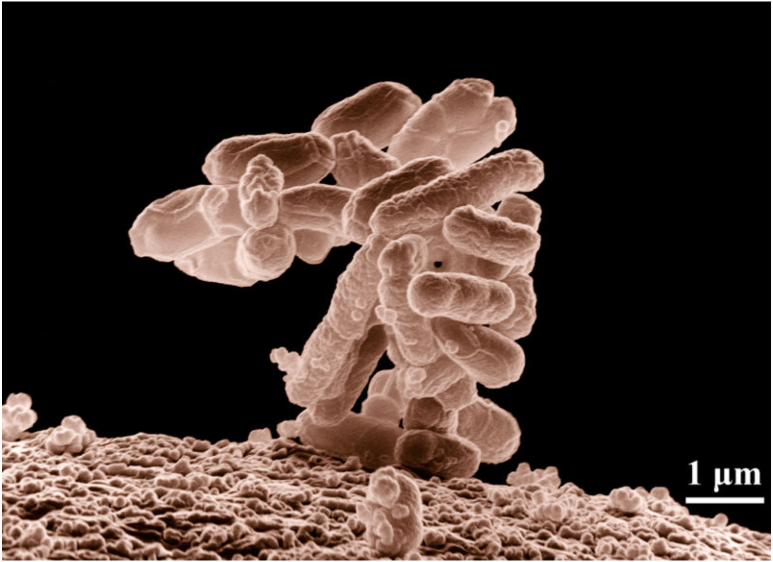
e.g.:

  - embryonic cycles in frog eggs

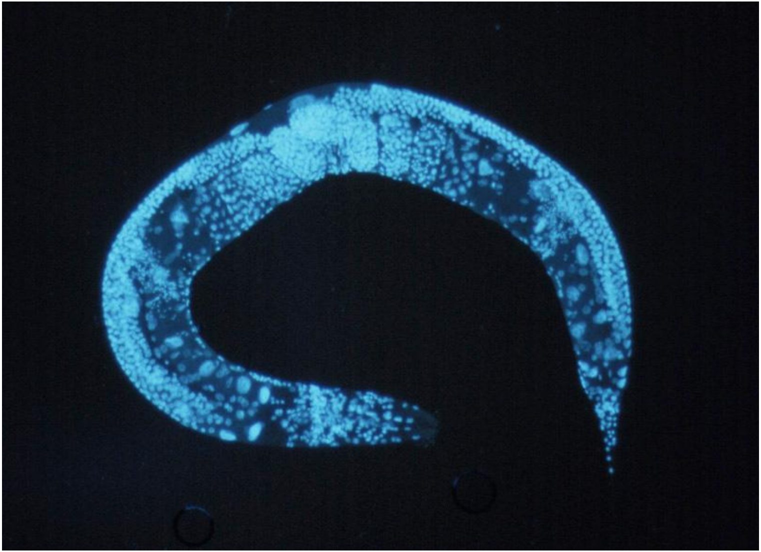
  - differentiation and development in flies

  - aging in worms

Ecoli



Caenorhabditis Elegans

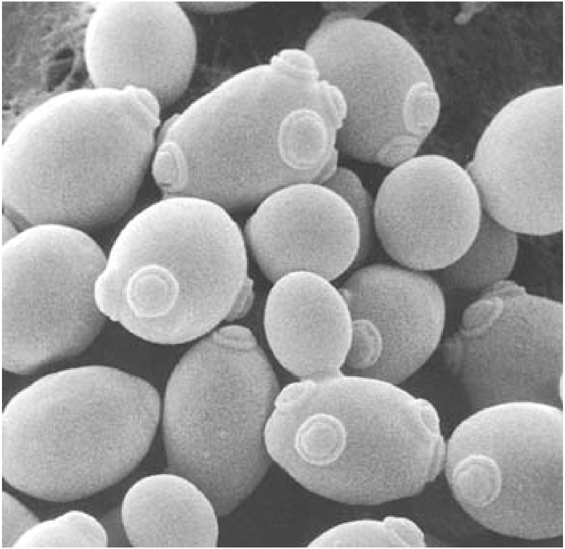


Drosophila Melanogaster

Mice

Xenopus (frog)

(Budding = Bakers's) Yeast



Arabidopsis Thaliana

Genomes Vary in Length

E.coli $\approx 5.44 \cdot 10^6$ bp, 5,400 genes

S.cerevisiae $\approx 1.2 \cdot 10^7$ bp, 5,800 genes

Drosophila $\approx 1.22 \cdot 10^8$ bp, 13,400 genes

C.elegans $\approx 10^8$ bp, 19,400 genes

Humans $\approx 3.3 \cdot 10^9$ bp, 20-25,000 genes

Arabidopsis $\approx 1.15 \cdot 10^8$ bp, 28,000 genes

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primary components of living things:

- sensors (*receptors*)

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(cell) life ≈ proteins and how/when they are produced

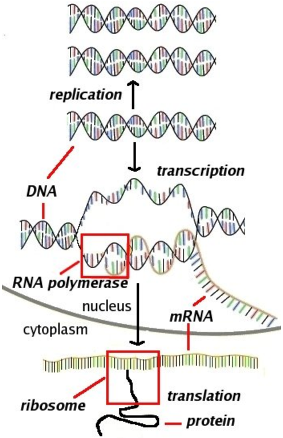
Gene Expression: DNA → Working Proteins

genome: genetic information of individual in DNA
"parts list" for proteins *(potentially)* in each cell

central dogma of molecular biology:

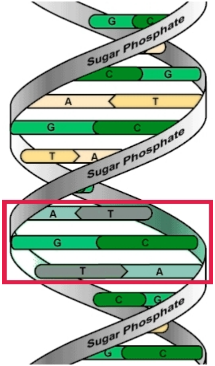
- DNA → RNA
- RNA → amino acid sequence
- folding into functional proteins

(also: replication at cell division)



(in biology, every "theorem" has exceptions: intermediate editing of RNA & proteins; RNA interference; prions, . . .)

DNA: "Hard Disk" Storage



deoxyribonucleic acid (DNA) molecules:
sequences of nucleotides $\in \{A, T, C, G\}$
(Adenine/Thymine/Cytosine/Guanine)
double-stranded helix
w/complementary strands: A-T, C-G

held together by hydrogen bonds

stable, good long-term memory, but
can break-up for replication/transcription

described as
"`gcacgagtaaacatgcacttcccaggccacagcagcaagaa...`"

genes = substrings coding for proteins

(also: regulatory and start/stop regions;
& regions coding for RNA that may not lead to proteins)

Transcription: Data, Progs $\rightarrow$ Working Memory

first step in gene expression: read-out genetic information:

DNA $\rightsquigarrow$ (messenger)RNA, transcribed letter by letter

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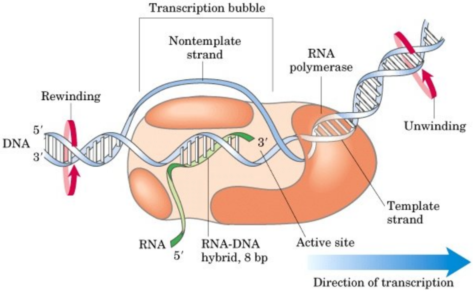
ribonucleic acid (RNA): similar to DNA, but
usually single-stranded; less stable, destroyed after use

RNA language same as DNA's (except $T \rightarrow U$, uracil)

RNA polymerase
"copying machine"



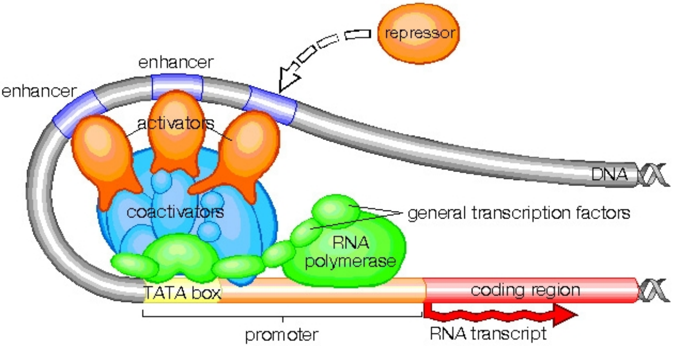
(in eukaryotic cells, "editing" gets rid of "introns"...)

Transcriptional Control Mechanisms

promoter region: DNA subsequence recognized by RNAP

in prokaryotes: short sequences, -35 and -10 nucleotides < gene

in eukaryotes: more sophisticated; need to time gene activity
in particular cells, or different tissues



distant enhancers, silencers, etc may also regulate promoters

Translation: Assembling the Protein

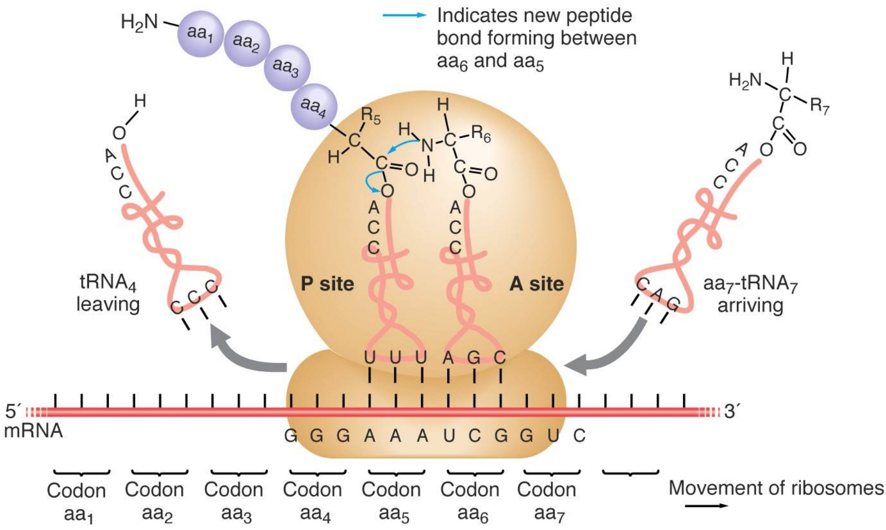
genetic code: $4^3 = 64$ codons (triplets) $\rightarrow$ 20 aminoacids + 2 start/stop

UUU UUC phenyl alanine UUA UUG leucine	UCU UCC UCA UCG serine	UAU UAC tyrosine UAA UAG stop	UGU UGC cysteine UGA stop UGG tryptophan
CUU CUC CUA CUG leucine	CCU CCC CCA CCG proline	CAU CAC histidine CAA CAG glutamine	CGU CGC CGA CGG arginine
AUU AUC AUA isoleucine AUG methionine	ACU ACC ACA ACG threonine	AAU AAC asparagine AAA AAG lysine	AGU AGC serine AGA AGG arginine
GUU GUC GUA GUG valine	GCU GCC GCA GCG alanine	GAU GAC aspartic acid GAA GAG glutamic acid	GGU GGC GGA GGG glycine

e.g.: DNA string *TACTCATTG* transcribed into
RNA string *AUGAGUAAC* (complement, & replace $T \rightarrow U$)
and then translated into Methionine-Serine-Asparagine

$\boxed{\textit{\& protein ``folds'' into particular 3D shape}}$

Ribosomes


Indicates new peptide bond forming between aa6 and aa5
H2N
aa1
aa2
aa3
aa4
R5
H
C
C=O
O
A
C
C
H
N
H
C
R6
C=O
O
A
C
C
P site
A site
tRNA4 leaving
aa7-tRNA7 arriving
H2N
C
R7
C
O
O
A
C
C
C A G
C C C
U U U
A G C
5´
3´
mRNA
G G G A A A U C G G U C
Codon aa1
Codon aa2
Codon aa3
Codon aa4
Codon aa5
Codon aa6
Codon aa7
Movement of ribosomes

Proteomics / Structural Biology: 3D Shape?

shape of proteins largely determine function:

proteins interact (w/each other, DNA, ligands, metabolites)

and fitting depends on amino acids appropriately placed

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- physical (X-ray crystallography, NMR spectroscopy)

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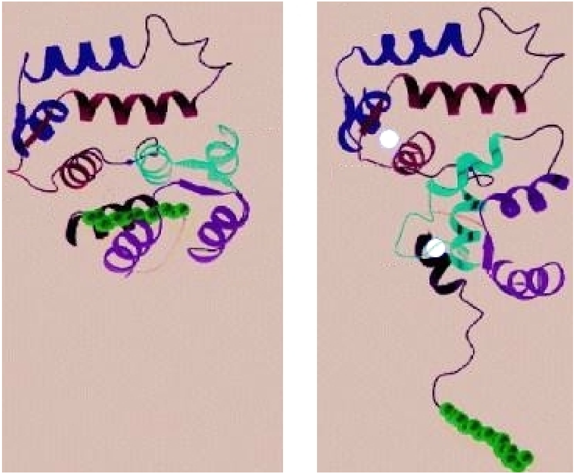
- optimization (hard: appropriate potential=?)
- physical (X-ray crystallography, NMR spectroscopy)
- bioinformatics (dist(gene_1,gene_2) $\approx 0 \Rightarrow$ "homologous")

"probably share evolutionary ancestry, so similar function"

"distance": (weighted) Hamming; count insertions/deletions

Protein Shape can be Controlled

conformation modified by interactions w/other molecules



example:

recoverin protein

(in retina)

Ca^{2+} inserted $\Rightarrow$

"arm" swings out

such modifications allow for signaling detection and relays

Sensors: Receptors

sense hormones, neurotransmitters, metabolites, . . .

& relay information to cell interior

reactions in cell triggered by ligand binding

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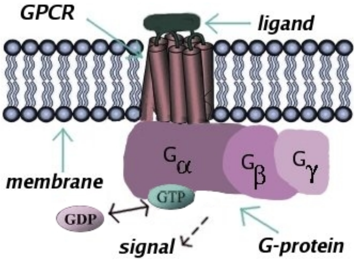

e.g. *G-protein-coupled receptors:*

common target of pharmaceuticals

binding $\Rightarrow$ conformation change

$\Rightarrow$ activate "G-proteins"

(*guanine triphosphate* (GTP)

& *diphosphate* (GDP) activation)

ligand binding starts/modulates activity of cell pathways

Relaying of Signals: Phosphorylation

$\boxed{\text{detect signals / \textit{process and relay} / actuate}}$

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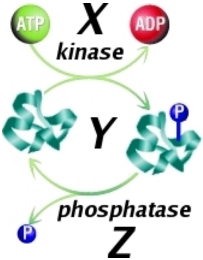
kinase (enzyme) X helps transfer phosphate group to Y

e.g. from "energy currency" *adenosine triphosphate (ATP)*



Y "activated" $\Rightarrow$ itself may act as kinase

deactivation: *phosphatase* Z removes P

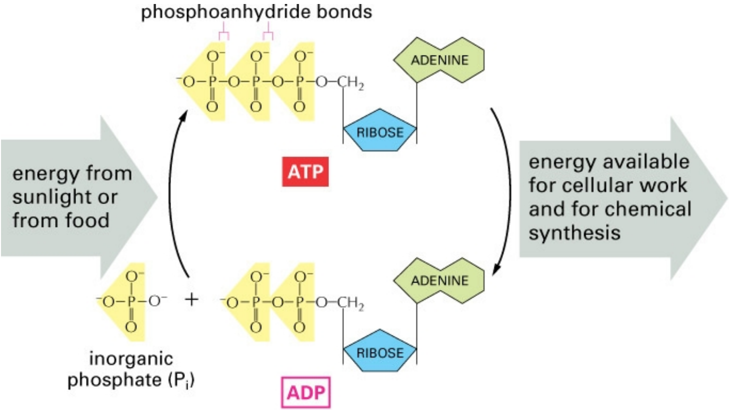
$\therefore$ signaling turned off & ready to detect new

often cascade $X \rightarrow Y \rightarrow \ldots \rightarrow$ *transcription factor*

failures to turn-off in cell-division-instructing $\Rightarrow$ cancer

adenosine triphosphate nucleotide is “energy currency”

figure 3.2 from Essential Cell Biology, Second Edition, published by Garland Science in 2004; ©by Alberts et al:



usually recycled in mitochondria where it is “recharged” into ATP
[chloroplasts in plants; cell wall or cytosol (glycolysis), in prokaryotes]
human cells contain ≈ one billion ATP molecules,
only sufficient for a few minutes

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- RNA interference (RNAi)

recently discovered control mechanism,
controls transcription by cleaving targeted mRNA's

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- epigenetic changes: histone modification, DNA methylation, ...

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Fluorescent Proteins

"at what rate is gene X is being transcribed?"

(under cell conditions being studied)

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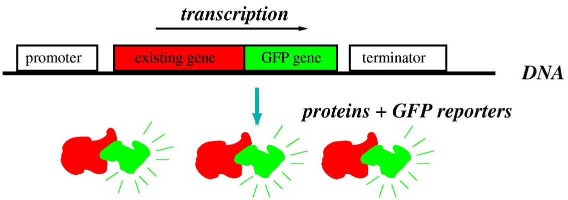
green [yellow . . .] *fluorescent protein (GFP)*: fluoresces under UV

GFP gene is inserted (cloned) into chromosome [or plasmid]

adjacent to or very close to the location of gene X

so, both controlled by same promoter region

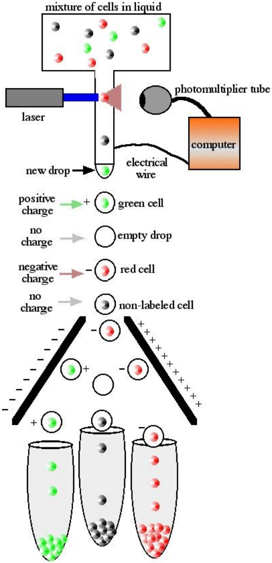


X & GFP transcribed/translated expressed simultaneously

measure intensity of GFP light $\Rightarrow$ X expression level

Fluorescence Activated Cell Sorting (FACS)

GFP particularly useful combined with *flow cytometry*:



sort cells into groups

based upon

cell size,

shape,

fluorescence, ...

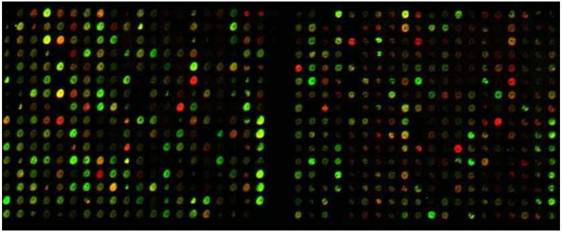
(thousands of cells/sec)

permits study of stochasticity

(individual expression)

Gene Arrays

(DNA chips, DNA microarrays, Affymatrix gene chips, ...)
allow "snapshot" of gene expression activity in cell



built using robotics & imaging, $\sim$ electronic chip fab

at (i, j), detector "tuned" to gene or nucleotide sequence X_{ij} has complement $\overline{X}_{ij}$ of X_{ij} (reverse-transcribed cDNA)

X_{ij} "sticks" to $\overline{X}_{ij}$ (hybridization: A-T, G-C base pairings)

X_{ij}'s radioactively or fluorescently tagged so can read-out

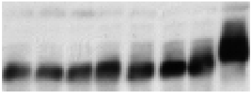
$\rightsquigarrow$ co-expression gene clustering; data for identification

Western Blots

method for detecting specific (forms of) proteins:

place extract on membranes w/antibodies that recognize specific proteins

visualize the results using e.g. radioactive labeling of stains



relative abundance of phosphorylated/un-phosphorylated X

in several experiments (higher placement: P)

("Southern" for DNA, "Northern" for RNA)

Limitations (!)

intrinsically noisy measurements:

chemical interactions in blots,

production errors in arrays,

. . .

and very few bits of information

pointless to try to tightly fit model parameters to such data

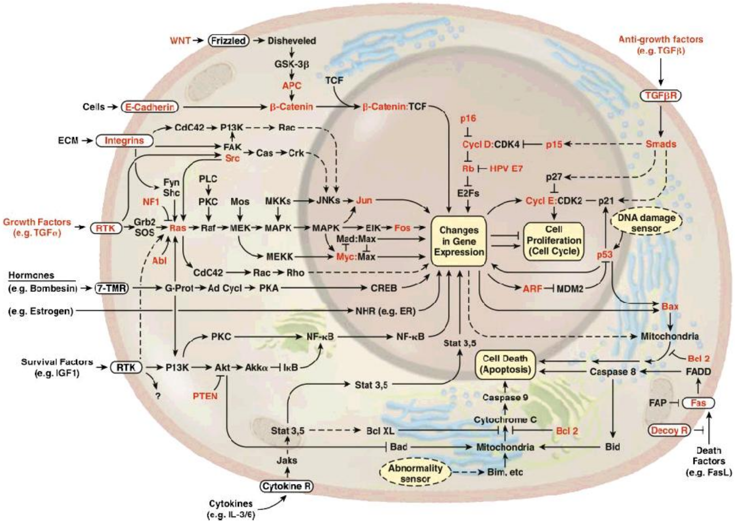
although *some* types of quantities

(channel currents, certain enzymes & metabolites)

can be measured far more accurately

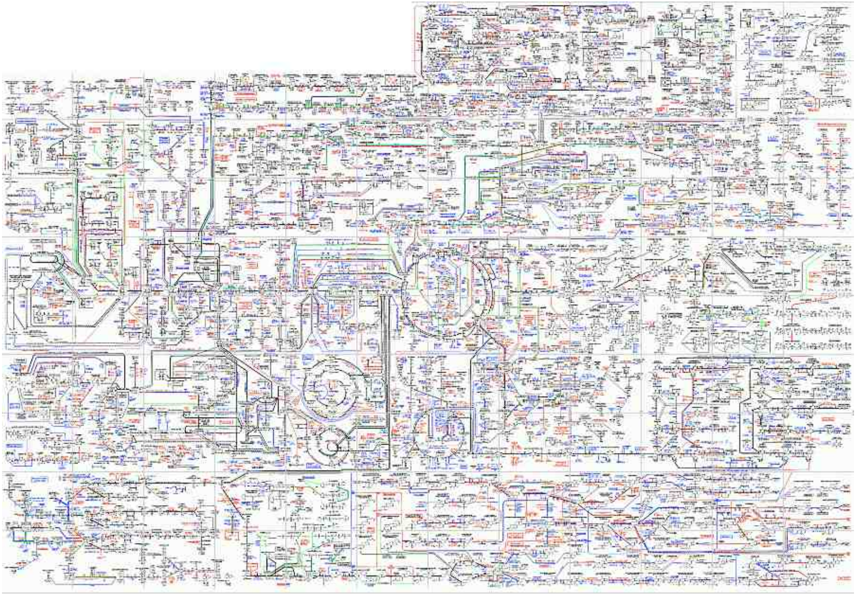
Outline

Cell Signaling / Gene Expression Example: Cancer Network

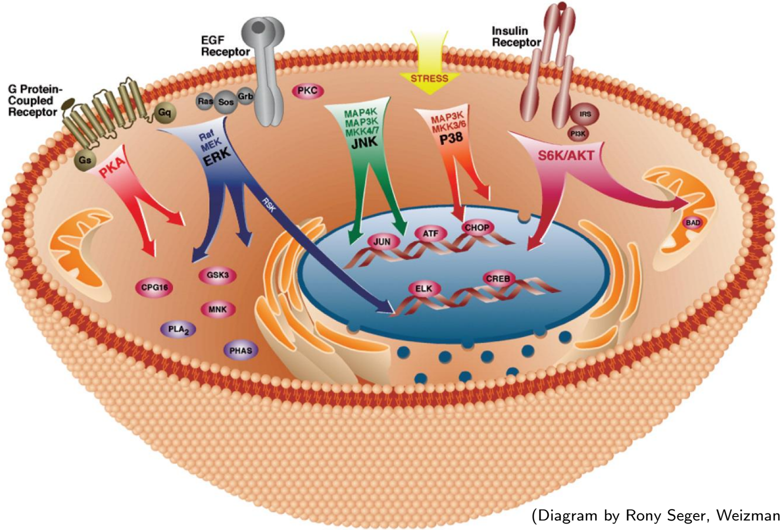


The hallmarks of cancer, Hanahan & Weinberg, Cell 2000

or. . . Metabolic Pathways



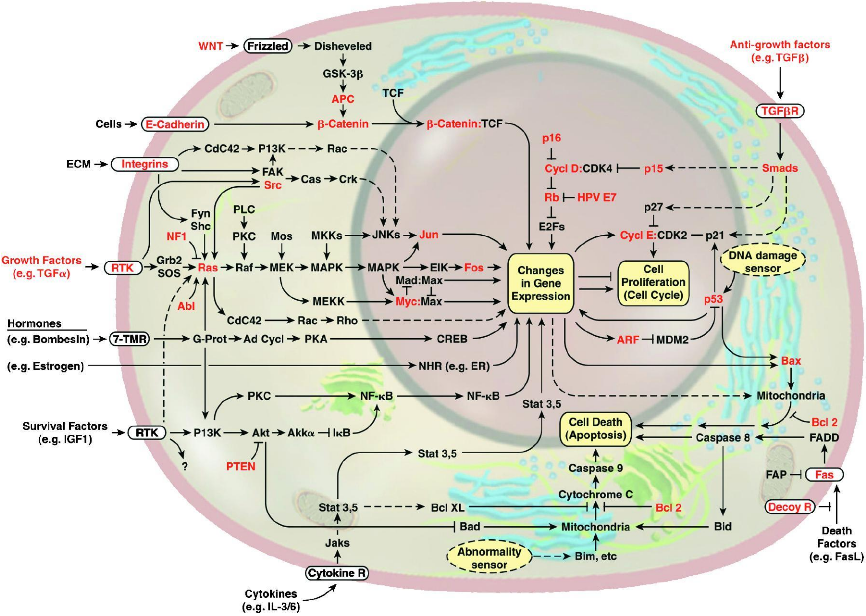
e.g. module: **M**itogen-**A**ctivated **P**rotein **K**inase cascades



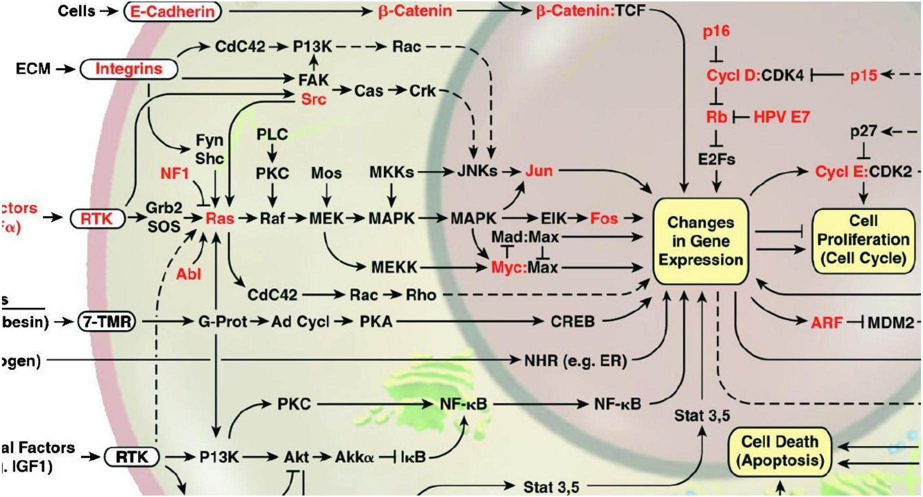
(Diagram by Rony Seger, Weizmann)

ubiquitous signaling “submodule” in eukaryotes

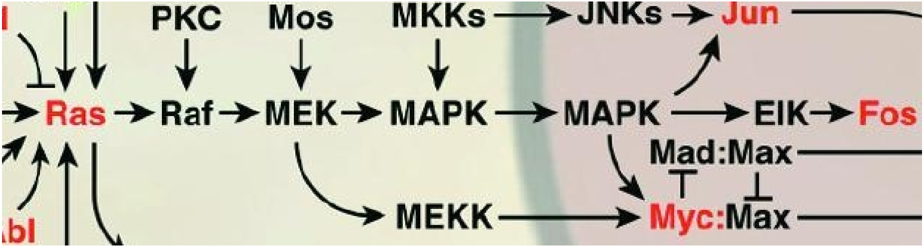
involved in proliferation, differentiation, movement, apoptosis, . . .



WNT
Frizzled
Disheveled
GSK-3β
APC
TCF
Cells
E-Cadherin
β-Catenin
β-Catenin:TCF
Anti-growth factors
(e.g. TGFβ)
TGFβR
Smads
ECM
Integrins
CdC42
P13K
Rac
FAK
Cas
Crk
Src
Fyn
Shc
PLC
PKC
Mos
MKKs
JNKs
Jun
NF1
p16
Cycl D:CDK4
p15
Rb
HPV E7
E2Fs
p27
Cycl E:CDK2
p21
DNA damage
sensor
Growth Factors
(e.g. TGFα)
RTK
Grb2
SOS
Ras
Raf
MEK
MAPK
MAPK
ElK
Fos
Mad:Max
Myc:Max
MEKK
Changes
in Gene
Expression
Cell
Proliferation
(Cell Cycle)
p53
Abl
CdC42
Rac
Rho
Hormones
(e.g. Bombesin)
7-TMR
G-Prot
Ad Cycl
PKA
CREB
ARF
MDM2
(e.g. Estrogen)
NHR (e.g. ER)
Bax
PKC
NF-κB
NF-κB
Stat 3,5
Mitochondria
Survival Factors
(e.g. IGF1)
RTK
P13K
Akt
Akkα
IκB
?
PTEN
Cell Death
(Apoptosis)
Bcl 2
Caspase 8
FADD
Stat 3,5
Caspase 9
FAP
Fas
Cytochrome C
Stat 3,5
Bcl XL
Bcl 2
Decoy R
Bad
Mitochondria
Bid
Jaks
Abnormality
sensor
Bim, etc
Death
Factors
(e.g. FasL)
Cytokine R
Cytokines
(e.g. IL-3/6)



Cells
E-Cadherin
β-Catenin
β-Catenin:TCF
ECM
Integrins
CdC42
P13K
Rac
FAK
Src
Cas
Crk
p16
Cycl D:CDK4
p15
Rb
HPV E7
E2Fs
p27
Fyn
Shc
PLC
PKC
Mos
MKKs
JNKs
Jun
NF1
Cycl E:CDK2
ctors
Fα)
RTK
Grb2
SOS
Ras
Raf
MEK
MAPK
MAPK
ElK
Fos
Changes in Gene Expression
Cell Proliferation (Cell Cycle)
Mad:Max
Myc:Max
MEKK
Abl
CdC42
Rac
Rho
s
besin)
7-TMR
G-Prot
Ad Cycl
PKA
CREB
ARF
MDM2
ogen)
NHR (e.g. ER)
PKC
NF-κB
NF-κB
Stat 3,5
al Factors
. IGF1)
RTK
P13K
Akt
Akkα
IκB
Cell Death (Apoptosis)
Stat 3,5


PKC
Mos
MKKs
JNKs
Jun
Ras
Raf
MEK
MAPK
MAPK
ElK
Fos
Mad:Max
MEKK
Myc:Max

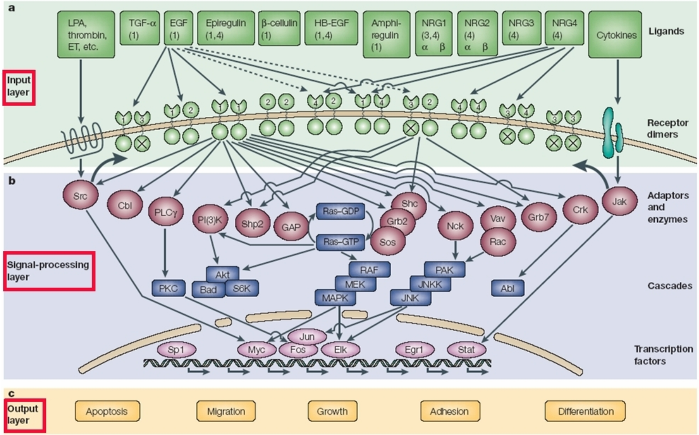
Signaling Cascade Animation

ANIMATION

animation from from biocreations.com

WEB ADDRESS

Inputs/States/Outputs; e.g.: ErbB



Yarden & Sliwkowski, Nature Rev Mol Cell Biol. 2001

inputs: hormones, neurotransmitters, lymphokines, . . .

conserved RTK pathway, EGF-family ligands

Herceptin® antibody blocks ErbB2 (breast cancer)

Natural Questions:

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  - identification (parameters = reaction constants)?

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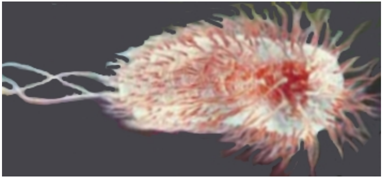
A Prokaryotic Signaling Pathway

chemotaxis: move in response to chemoattractants/repellants

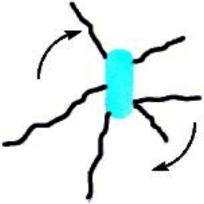


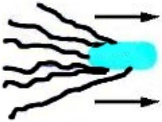
studied extensively in *E. coli*

single-celled, $\approx 2\ \mu$m;

flagella for movement

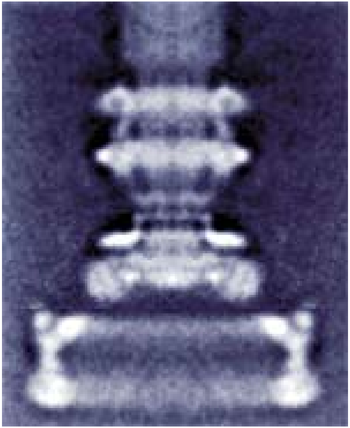
two modes: "tumble" (CW) or "run" (CCW $\Rightarrow$ bundle)

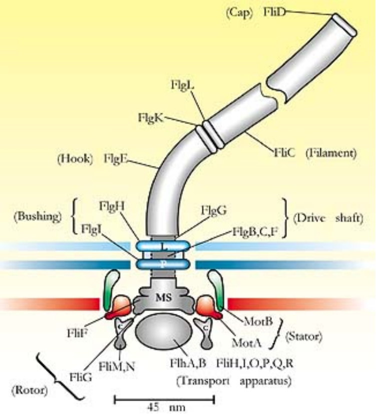




Typical trails

Flagellar Motors a Nanotechnological Wonder

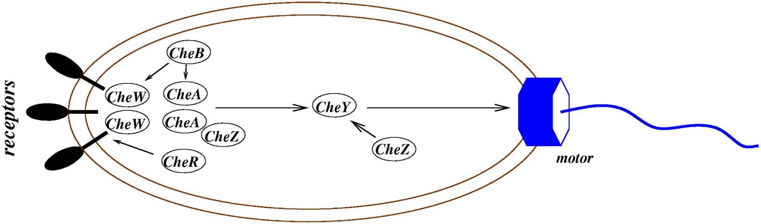




"engines, propellers, particle counters, rate meters, gear boxes"

(Howard Berg)

Signaling Pathways for *E. coli* Chemotaxis



CheY-P binds to motor $\Rightarrow$ tumble; else run (…to nutrient)

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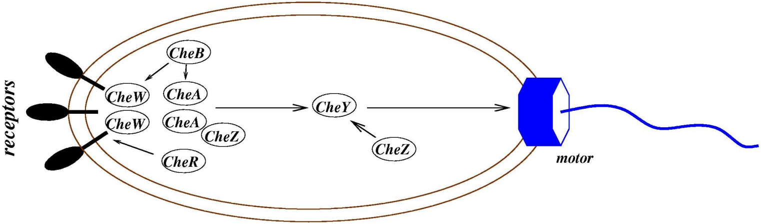

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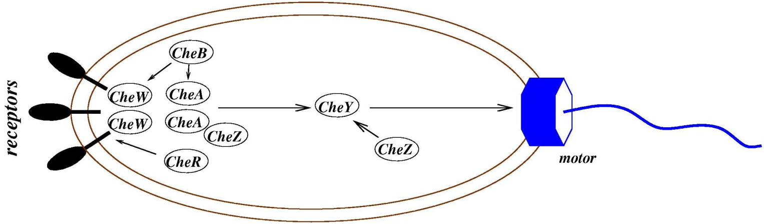

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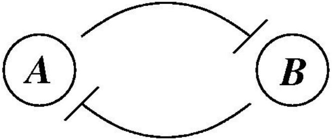
CheR/CheB feedback loop helps adapt to constants ($\nabla = 0$)
(integral feedback)

stochastic gradient search in a nutrient-potential landscape

Outline

Memory (Flip-Flop): "Genetic Toggle Switch" (J. Collins)



two repressible promoters,
mutually inhibiting

Gardner et. al., Nature 2000; Kobayashi et. al., PNAS 2004

two stable steady-states:

one repressor dominant, at expression level $\gg$ basal rate;
other at basal

switch of dominance induced by:

- signal that causes rapid decay of dominant
- or signal that boosts basal production rate of dominated

new steady state remains until next transition input signal

Typical Components

lacI gene / LacR protein (bacterial)

lactose metabolism, represses lac operon

λ*cI* gene / λ-repressor protein (bacteriophage λ)

promotes incorporation into host genome (lysogeny)

possible inputs:

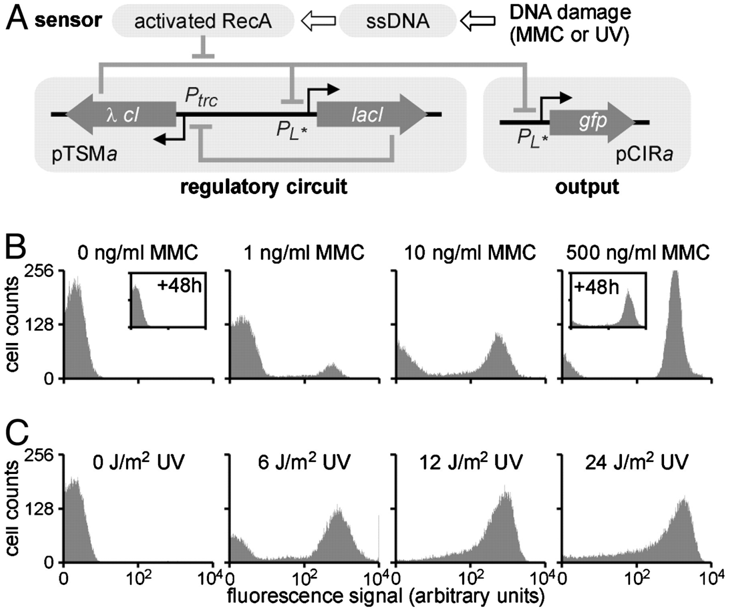
IPTG interferes w/ LacR repression

SOS-response pathway: RecA activated if single-stranded DNA,

& cleaves λ-repressor protein

DNA damage due to MMC (mitomycin C) or UV flips toggle

Use GFP Expression as Reporter



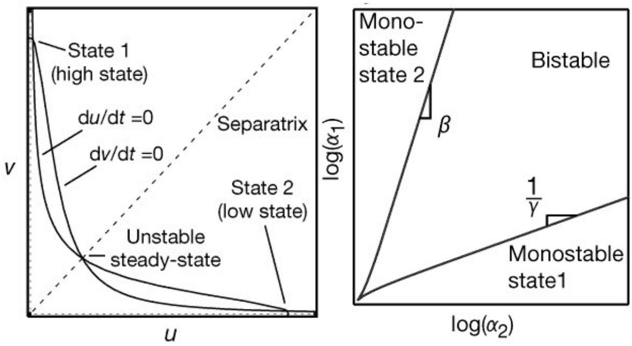
Model

$$
\begin{aligned}
(LacI \text{ mRNA}) \qquad \frac{du}{dt} &= \epsilon_1 \left(\alpha_{u,0} + \frac{\alpha_u}{1 + V^{\gamma_v}} \right) - u \\
(\lambda CI \text{ mRNA}) \qquad \frac{dv}{dt} &= \epsilon_1 \left(\alpha_{v,0} + \frac{\alpha_v}{1 + U^{\gamma_u}} \right) - v \\
(LacI \text{ protein}) \qquad \frac{dU}{dt} &= \zeta_u (u - U) \\
(\lambda CI \text{ protein}) \qquad \frac{dV}{dt} &= \zeta_v (v - V) \\
(\text{GFP}) \qquad \frac{dG}{dt} &= \epsilon_2 \left(\alpha_{G,0} + \frac{\alpha_G}{1 + U^{\gamma_u}} \right) - \delta_G G
\end{aligned}
$$

Variable	Description
u	Concentration of Repressor *LacI* mRNA
v	Concentration of Repressor λCI mRNA
U	Concentration of Repressor *LacI* Protein
V	Concentration of Repressor λCI Protein
G	Concentration of Green Fluorescent Protein (GFP)
$\alpha_{i,0}$	Basal Synthesis Rate of mRNA from gene i
$\alpha_{i,0} + \alpha_i$	Maximal Synthesis Rate of mRNA from gene i
γ_i	Hill Coefficient of cooperativity exhibited by protein i
ζ_i	Ratio of mRNA to protein lifetimes for gene i
ϵ_i	Plasmid Copy Number for plasmid i

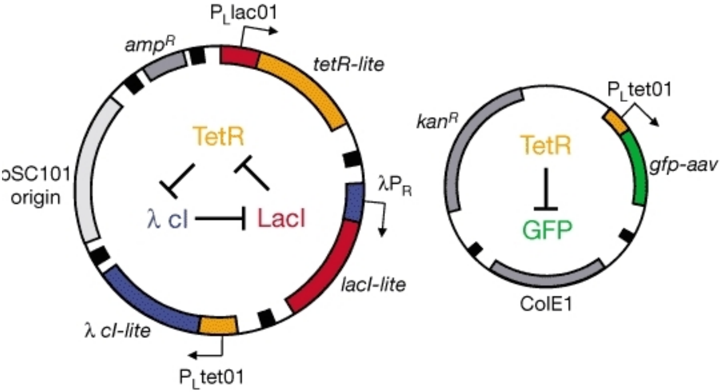
Reduced Model and Phase-Planes

$$\frac{du}{dt} = \frac{\alpha_1}{1 + v^\beta} - u \,, \qquad \frac{dv}{dt} = \frac{\alpha_2}{1 + u^\gamma} - v,$$



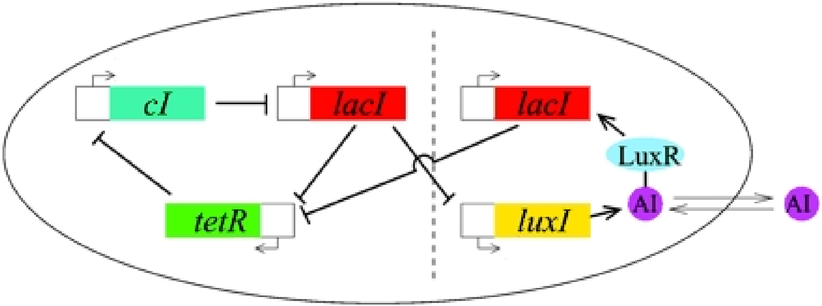
cooperativity (Hill coefficient) enlarges region of stability

Repressilator (Elowitz-Leibler) Oscillator



repressilator: lacI / TetR / cI (*E. coli*; last from λ-phage)

Intercell Communication (e.g., for Oscillator Synch)



Garcia-Ojalvo, Elowitz, Strogatz, PNAS'04

repressilator coupled to quorum-sensing module
uses LuxI, AutoInducer, LuxR
(from *Vibrio fischeri* bioluminescent marine bacterium)
AI concentration high $\Rightarrow$ binds LuxR, activates luciferase promoter

System Integration